

Software, translators, virus and cutting-edge technology

★ Golden

One college period · 40–45 minutes · Overview of Computer System

What you will learn

- Distinguish system, application and utility software
- Explain compiler, interpreter, assembler and compare compiler with interpreter
- Define computer virus and name antiviruses
- Outline AI and cloud computing at Grade XI level

Software. A set of programs, procedures and documentation that instruct the hardware.

Type	Role	Examples
System software	Runs the computer; manages resources	OS (Windows, Linux, macOS, Android), device drivers, BIOS
Language translator	Converts source code to machine code	Compiler, interpreter, assembler
Application software	User jobs	MS Word, Excel, Chrome, Turbo C IDE, Photoshop
Utility	Maintenance	Antivirus, compression (WinRAR), disk cleanup, backup, data recovery

Translators

	Compiler	Interpreter
How	Translates the whole program first, then runs the object code	Translates and runs one statement at a time
Speed	Faster at run time	Slower at run time
Errors	Shows a list of errors after compilation	Stops at the first error
Object file	Creates .exe / .obj	Usually no separate object file
Example	C, C++	Python, BASIC (classic)

Assembler translates assembly language (mnemonics) into machine code. Source code is the program you write; object code is the machine-language result.

Computer virus

LEARN IT

A computer virus is a malicious program that attaches itself to other files, replicates, and can damage data, steal information or slow the system. Worms spread on networks without a host file. Trojans pretend to be useful software. Ransomware locks files until a ransom is paid.

- Examples of viruses/malware: ILOVEYOU, WannaCry, Trojan horses, spyware, adware.
- Antiviruses: Kaspersky, Avast, AVG, Norton, Bitdefender, Windows Defender, McAfee, Avira.

Cutting-edge technology (Unit 1.5)

- Artificial Intelligence (AI): making machines perform tasks that need human intelligence — speech, vision, expert advice. Applications: face unlock, chatbots, medical imaging, self-driving research.
- Cloud computing: using servers on the internet (Google Drive, Microsoft 365) so you need less local storage. Advantages: access anywhere, backup, pay-as-you-go, easy sharing.

HOW BIEK ASKS THIS

2026 Paper I: define virus with examples and name five antiviruses; compiler vs interpreter; HTML as the markup language for web pages.

COMMON BOARD MISTAKE

Windows is system software; MS Word is application software. A virus is software, not hardware. HTML is a markup language, not a full programming language like C, but BIEK MCQs still key it as the language of web pages.

اردو میں سمجھیں

سافٹ ویئر پروگراموں کا مجموعہ ہے۔ سسٹم سافٹ ویئر کمپیوٹر چلاتا ہے، ایپلی کیشن صارف کا کام کرتی ہے، یوٹیلیٹی صفائی اور حفاظت کرتی ہے۔

Compiler پورا پروگرام ایک ساتھ مشین کوڈ بناتا ہے، Interpreter لائن بہ لائن وائرس نقصان دہ پروگرام ہے جو نقل کرتا ہے؛ اینٹی وائرس اس سے بچاتا ہے۔

CHECK YOURSELF

1. Name three language translators.
2. Give two advantages of cloud computing.

Answers: 1. Compiler, interpreter, assembler · 2. Access from anywhere; automatic backup / less local hardware

MCQs

1. A computer virus is a:
 - a. Signal
 - b. Software
 - c. Hardware
 - d. Firmware
2. Windows is an example of:
 - a. Application software
 - b. System software
 - c. Utility only

d. Compiler

3. The markup language used to produce web pages is:

- a. XML
- b. HTML
- c. SGML
- d. WML

4. C programs are converted into machine language by a:

- a. Editor
- b. Assembler
- c. Compiler
- d. Debugger

Key: 1=b, 2=b, 3=b, 4=c

Short questions (3 marks each)

1. How is a compiler more efficient than an interpreter?
2. Define computer virus with examples. Name any five antiviruses.
3. Differentiate system software and application software.

Model short answers

1. A compiler translates the whole program once into object code which then runs fast. An interpreter translates every line each time the program runs, so it is slower and stops at the first error.
2. A virus is malicious replicating software (examples: worms, Trojans, ransomware). Antiviruses: Kaspersky, Avast, Norton, Bitdefender, Windows Defender.
3. System software controls hardware (OS, drivers). Application software does user tasks (Word, browser, Turbo C).

Long questions (board Section C)

1. Explain types of software. Discuss language translators with a comparison of compiler and interpreter.